

Architecture





About the Course

The focus of the course in this rotation is on answering the question, "Why do buildings stand up?" The main text of that name offers the history of great monuments and the architectural technology that makes them possible. Famous buildings from 1800-1900 and 100 BC-350 AD are mentioned as well.



Placement & Combining Tips

Suitable for Grade 8 students with minimal adjustment if combining.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Architecture: Grades 9-12 1 time/week 30 min	Why Buildings Stand Up 50 Buildings You Should Know

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Architecture: Grades 9-12				
				Architecture: Grades 9-12



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Architecture: Grades 9-12



Why Buildings Stand Up



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)
(No Subject Supplies Assigned)



Quick Links

(No Quick Links Assigned)

[Click THIS text
or scan the QR
code for links.](#)



Architecture: Grades 9-12

How To Approach



Recap

Connect back to the previous lesson. Take a minute (without looking ideally) to recall the ideas you read about last week.



Read

Read the assigned selection.



View

Look at pictures from links or in the text of design features and/or famous buildings.



Narrate

Various ways to narrate are assigned in the lessons. Because architecture is an especially visual subject, try to incorporate sketches frequently into your narrations and thoughts.



Discuss

Discuss what you have learned and found interesting with a parent, teacher, or another student.



Connect

Plan time at the end of lesson or later in the afternoon to make additional drawings of buildings or features in a sketchbook and/or Book of Centuries.



Teacher Note

These lessons are written directly to the student, but to assist them throughout the year, please read through this section and the lesson plans so you are familiar with the instructions and flow of the lessons. Accountability components are added to facilitate teacher/student discussion and growing independence.

It is valuable for those in a classroom setting to also have a mix of lessons that students complete silently themselves and others read aloud and narrated together. Because of the visual elements of the course, it is especially suited to completing as a group. Classroom settings lend themselves well to additional discussion and sharing at the end of lessons after everyone has narrated either together or independently with their notebooks.

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Term 1

WEEK 1 30m Architecture: Grades 9-12 - Lesson 1

Structures

Materials: Why Buildings Stand Up, notebook

→ INTRO

Glance through the Contents page so you have a sense of what this book is going to be about. We don't often stop to think about why buildings stand up or consider all the different ways people have built in the past. As we learn about the science of construction, look around at the built structures in your area and see if you can begin to understand some of the secrets of their construction.

→ READ

Preface and Ch.1 p.17-26

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself. With your sketches and written work, you will be creating an architecture notebook that will explore why buildings stand up. Be as creative as you like and enjoy the adventure!

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, for this reading you might want to write about the author's ideas about structure and function in architecture. What examples can you think of or what did you think about his ideas or what did his ideas make you ask questions about?

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts with other students, your teacher, your parent, or another adult.

WEEK 2 30m Architecture: Grades 9-12 - Lesson 2

The Pyramids

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.2 p.27-38

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Because you narrated the whole reading silently to yourself, you now have a few moments to focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

- On p.28, Salvadori says, "Cultures are immortalized by monuments,

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Term 1

which express their conception of the world, of life and death," what monuments today immortalize our conception of the world and of life and death as a culture?

WEEK 3 30m Architecture: Grades 9-12 - Lesson 3

The Pyramids, Loads

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.2 p.38-Ch.3 p.49

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the reading that was especially interesting to you or write about the different loads discussed.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts with other students, your teacher, your parent, or another adult.

• EXTRA HELPING

∞ Video Link: Decoding the Great Pyramid

WEEK 4 30m Architecture: Grades 9-12 - Lesson 4

Loads

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.3 p.49-58

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 5 30m Architecture: Grades 9-12 - Lesson 5

Materials

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.4 p.59-66

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you or that you thought was most important.

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→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 6 **30m Architecture: Grades 9-12 - Lesson 6**

Materials, Beams and Columns

 Materials: Why Buildings Stand Up, notebook

→ READ

Ch.4 p.66-Ch.5 p.76

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 7 **30m Architecture: Grades 9-12 - Lesson 7**

Beams and Columns

 Materials: Why Buildings Stand Up, notebook

→ READ

Ch.5 p.76-89

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 8 **30m Architecture: Grades 9-12 - Lesson 8**

Houses

 Materials: Why Buildings Stand Up, notebook

→ READ

Ch.6 p.90-101

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

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→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 9 30m Architecture: Grades 9-12 - Lesson 9

Houses, Skyscrapers

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.6 p.101-Ch.7 p.109

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 10 30m Architecture: Grades 9-12 - Lesson 10

Skyscrapers

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.7 p.109-116

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 11 30m Architecture: Grades 9-12 - Lesson 11

Skyscrapers

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.7 p.116-125

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

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Term 1

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 12 30m Architecture: Grades 9-12 - Lesson 12

Exam Week

→ EXAMS

Answer question(s) related to the course.

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Term 2

WEEK 1 30m Architecture: Grades 9-12 - Lesson 13

Eiffel Tower

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.8 p.126-133

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 2 30m Architecture: Grades 9-12 - Lesson 14

Eiffel Tower

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.8 p.133-143

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 3 30m Architecture: Grades 9-12 - Lesson 15

Bridges

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.9 p.144-157

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

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Term 2

WEEK 4 30m Architecture: Grades 9-12 - Lesson 16

Bridges

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.9 p.157-164

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 5 30m Architecture: Grades 9-12 - Lesson 17

Brooklyn Bridge

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.10 p.165-176

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving an overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

WEEK 6 30m Architecture: Grades 9-12 - Lesson 18

Brooklyn Bridge, Form-Resistant Structures

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.10 p.176-Ch.11 p.186

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of

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the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving an overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

WEEK 7 30m Architecture: Grades 9-12 - Lesson 19

Form-Resistant Structures

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.11 p.186-195

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving an overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

WEEK 8 30m Architecture: Grades 9-12 - Lesson 20

Form-Resistant Structures

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.11 p.195-205

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving an overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

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WEEK 9 30m Architecture: Grades 9-12 - Lesson 21

Unfinished Cathedral

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.12 p.206-217

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving an overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

WEEK 10 30m Architecture: Grades 9-12 - Lesson 22

Unfinished Cathedral

Materials: Why Buildings Stand Up, notebook

→ READ

Ch.12 p.217-224

→ NARRATE

Copy any diagrams or sketches that would be helpful to you in your notebook. Silently go over the reading in your mind. Sometimes it helps to talk aloud to yourself at first until you can silently narrate to yourself.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the chapter that was especially interesting to you. For example, what about the construction process of the pyramids was most impressive or interesting to you? Explain in detail one aspect of the process. Instead of giving a whole overview narration of the reading, focus in on one part and give greater detail about that part.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down or your answer to the following discussion prompt with other students, your teacher, your parent, or another adult.

WEEK 11 30m Architecture: Grades 9-12 - Lesson 23

La Sagrada Familia

Materials: 50 Buildings You Should Know (Optional), notebook

→ INTRO

If you have the book, *50 Buildings You Should Know*, we are going to look at Building 36. If you do not have it, that is fine. There is a link for a video to watch.

• EXTRA HELPING

∞ Video Link: Sagrada Familia Finally Nears Completion

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Term 2

→ READ

Building 36, p.112-113

→ WATCH

∞ Video Link: How Gaudi Designed the Sagrada Familia

→ SKETCH

Sketch something of this amazing building in your notebook.

→ WRITE

Use the remaining time to write at least a paragraph about one aspect of the La Sagrada Familia or about Gaudi and the story of his work that was especially interesting to you.

→ DISCUSS

Depending on your setting and schedule, discuss your thoughts that you wrote down with other students, your teacher, your parent, or another adult.

WEEK 12 30m Architecture: Grades 9-12 - Lesson 24

Exam Week

→ EXAMS

Answer question(s) related to the course.

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Term 3

WEEK 1 30m Architecture: Grades 9-12 - Lesson 25

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 2 30m Architecture: Grades 9-12 - Lesson 26

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 3 30m Architecture: Grades 9-12 - Lesson 27

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 4 30m Architecture: Grades 9-12 - Lesson 28

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 5 30m Architecture: Grades 9-12 - Lesson 29

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 6 30m Architecture: Grades 9-12 - Lesson 30

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

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Term 3

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 7 30m Architecture: Grades 9-12 - Lesson 31

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 8 30m Architecture: Grades 9-12 - Lesson 32

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 9 30m Architecture: Grades 9-12 - Lesson 33

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 10 30m Architecture: Grades 9-12 - Lesson 34

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

WEEK 11 30m Architecture: Grades 9-12 - Lesson 35

Materials: Why Buildings Stand Up, notebook

→ READ

~9 pages or what you can read in 20 min

→NARRATE: COPY ANY DIAGRAMS OR SKETCHES THAT ARE HELPFUL. NARRATE SILENTLY AND THEN WRITE ABOUT A SPECIFIC SECTION OR IDEA FROM THE READING.

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Term 3

WEEK 12  **30m Architecture: Grades 9-12 - Lesson 36**

Exam Week

→ **EXAMS**

Answer question(s) related to the course.

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Examination

Term 1

- Show your architecture notebook to your teacher. Write what you remember about pyramids, loads, materials, beams and columns, houses, or skyscrapers.